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Enhancing chronic migraine preventive therapy: low-level 810 nm laser acupuncture as an add-on treatment for patients with unsatisfactory pharmacological effect, a pilot single-blind randomized controlled trial

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Abstract

Background Laser acupuncture is a proven non-invasive treatment with effects comparable to traditional acupuncture in different types of headaches, but there is still insufficient evidence for chronic migraine (CM) in adults. We aim to investigate the efficacy and safety of laser acupuncture (LA) as an add-on preventive therapy on CM.

Methods A single-blind randomized controlled trial was conducted from January 2022 to November 2023. CM patients with unsatisfactory pharmacological effects were randomly assigned in a 1:1 ratio to receive either LA or sham treatment over a course of 8 sessions spanning 4 weeks. The co-primary outcomes were changes in monthly migraine days (MMD) and acute headache medications usage days per month from baseline. Evaluations were taken at baseline (12 weeks before randomization), at 4th week (treatment completed), 8th week and 12th week from baseline.

Results A total of 60 patients (30 in each group) were included in the intention-to-treat analyses. Baseline headache characteristics between trial groups were similar. Compared with the sham group, the LA group had a significant reduction in MMD (5.2 vs. 1.5 days at 8th week, $p=0.015$; 7.3 vs. 1.8 days at 12th week, $p=0.001$), and acute headache medications usage days per month (3.1 vs. 0.4 days at 4th week, $p=0.007$; 3.2 vs. 0.0 days at 8th week, $p=0.005$; 3.9 vs. 0.0 days at 12th week, $p<0.001$). No serious adverse event was observed in both groups.

Conclusions Laser acupuncture was effective in reducing MMD and acute headache medications usage with promising safety. Specifically, the efficacy of LA exhibited a progressively more pronounced effect within the follow-up period. We suggested that LA is a promising add-on preventive therapy for CM, and trials focused on investigating the mechanism of LA's effect and its long-term effects on CM prevention are justified.

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Trial registration The study was retrospectively registered at ISRCTN.org Identifier: ISRCTN11208146 (https://doi.org/10.1186/ISRCTN11208146). The registration date: 19, January, 2024.

The date of first participant registration: 04, May, 2022.

Keywords Migraine, Laser acupuncture, Prevention, Randomized controlled trial

Introduction

Among patients under the age of 50, migraine is currently the leading cause of disability [1] and the second highest cause overall [2]. The estimated global prevalence of migraine currently reaches 14–15%[2]. For migraine patients, the most debilitating condition is chronic migraine (CM), which is a major global economic burden due to healthcare costs and reduced job-related productivity [3]. Indeed, the total annual costs for patients with CM have ranged from EUR 6,443 to EUR 53,446 (GBP 2021), depending on the number of headache days per month [4].

The pathophysiology of migraine is complex, including cortical spreading depression, activation of the trigeminovascular pathway, peripheral and central sensitization, and releasing of neuropeptides such as calcitonin gene related peptide (CGRP) [5]. This process causes signaling along the trigeminovascular pathway, which diffuses and activates meningeal nociceptors. Ultimately, this results in hyper-responsiveness within primary afferent fibers and/or central neurons, known as peripheral sensitization, leading to the throbbing pain of migraine [5]. Additionally, the sensitization of central trigeminovascular neurons, known as central sensitization, causes cephalic and extracephalic allodynia [5].CM typically evolves from episodic migraine because of lowered thresholds in nociception, sensitization, and structural brain changes [6]. As CM pathophysiology attracts great attention, therapeutic strategies for CM patients have also advanced markedly over the past 5 years.

Despite the emerging of preventive treatments of CM, there are still some patients encounter ineffectiveness according to real-world studies [6]. In addition, as the duration of preventive treatment prolongs, many CM patients show unsatisfactory responses to treatments, including some experiencing a waning of effectiveness [7, 8]. In clinical practice, it is noted that these patients represent the majority, approximately two-thirds of patients using traditional preventive therapy [9]. And complementary and alternative therapies may serve as an add-on treatment in addition to conventional medications. Therefore, complementary, and alternative therapies have become indispensable preventive treatments for CM [10]. Among these therapies, acupuncture is known to be a suitable preventive treatment for migraine [11, 12].

Laser acupuncture (LA), a non-invasive, non-thermal photonic stimulation of acupoints, is a well-established treatment option with effects equivalent to traditional acupuncture as shown in several chronic pain diseases [13, 14], and the treatment of allergic rhinitis [15]. Although multiple studies have reported the therapeutic effects of acupuncture in migraine prevention [11, 12], the evidence for LA in CM prevention is limited. As far as we know, LA has fewer adverse effects in treating different types of headaches, like chronic tension-type headache in adults [16] and migraine in children [17]. In fact, participants in these two studies who received LA did not report any side effects [16, 17]. However, for CM in adults, there is only one older study with a small sample size ($n=12$) and non-specific headache types (both migraine and muscle tension headache) similar to the present one [18]. This study revealed inconclusive results for the application of LA to CM in adults.

We hypothesized that LA enhances the pharmacological treatment effect as an add-on preventive therapy for individuals with poor-controlled CM. Therefore, the aim of our study is to investigate the efficacy and safety of LA as an add-on preventive therapy on CM managed pharmacologically with unsatisfactory effect.

Methods

Trial design and participants

Our study is a hospital-based (Taichung Veterans General Hospital, which serves mainly Han-Chinese population, with Fitzpatrick scale skin type III to IV), single-blind randomized controlled trial. Between January 2022 to November 2023, patients over 20 years old with CM managed pharmacologically with unsatisfactory effect were included. CM was diagnosed by neurologists based on the criteria of International Classification of Headache Disorders, 3rd edition [19]. Participants were randomly assigned in a 1:1 ratio to receive either LA or sham treatment. The flow diagram was shown in Fig. 1.

Participants were randomized by computer software (Excel 2016 for Windows) and the randomization process was overseen by the Institutional Review Board at Taichung Veterans General Hospital. All the participants and our collaborative case-manager were blinded to treatment types. Clinical assessments were taken at baseline, with headache defined in diary for 12 weeks before randomization, and other clinical assessments at

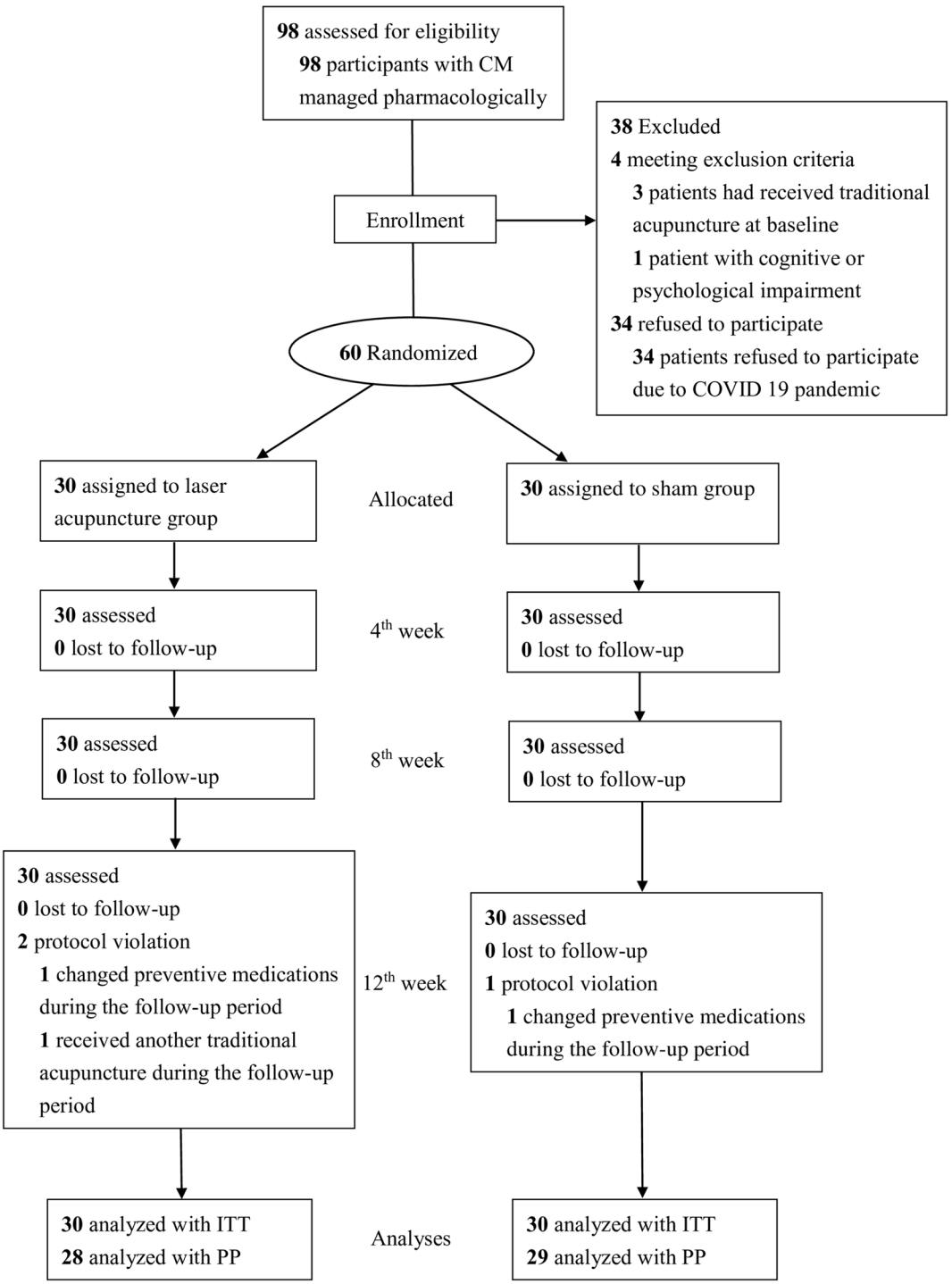


Fig. 1 Flow diagram. ITT: intention-to-treat; PP: per-protocol

the time of randomization, at the 4th (treatment completed), 8th and 12th week from baseline. After completing the LA protocol or sham treatment, participants were observed via clinical follow-up examinations and through questionnaires. In addition, participants received anthropometric measurements, including body-mass index (BMI) by case-manager at the time of randomization. Medication histories of patients were recorded, including acute headache medications and preventive medications. The participants' medication

histories were obtained from medical records at Taichung Veterans General Hospital.

The clinical assessments protocol includes the following two steps: (1) the headache diary procedure was completed by each participant and the data were collected by the case manager. Items recorded included headache lasting time, frequency, location, severity (using the Numeric Rating Scale [NRS], a determination of pain level on a scale from 0 to 10), acute headache medication usage, presence of aura, menstrual relation, and concomitant symptoms; (2) the MIDAS questionnaires were administered under the supervision of a trained and certified case manager, who ensured that the participant understood each question. The case manager was blinded to treatment types and certified by neurologists.

Trial inclusion and exclusion criteria

The inclusion criteria included: (1) patients aged > 20 years old and had CM managed pharmacologically, including preventive and/or acute migraine medications, and in addition, those who had refused preventive agent despite recommendation of the neurologist; (2) patients who had unsatisfactory effect of current pharmacological treatments, defined as patient self-reported or direct questions asked by a neurologist (e.g., "Are you comfortable enough with medication for CM?") [20]; and (3) patients who had a minimum of one-year history of migraine with or without aura.

Exclusion criteria included: (1) patients who had received another LA therapy or traditional acupuncture at baseline; (2) headache onset after the age of 50; (3) cognitive or psychological impairment interfering with the participant's ability to receive LA protocol and describe symptoms, which is defined by the determination of neurologist; and (4) patients with missing data at baseline.

In our trial, the primary analytical approach utilized is intention-to-treat (ITT) analyses. Simultaneously, we have also performed per-protocol (PP) analyses. The factors listed below was defined as protocol violations which would be excluded from the per-protocol analysis: (1) patients who had records of changing preventive medications during the follow-up period, defined as any changes in all preventive medications during the study period; (2) patients who received another LA therapy or traditional acupuncture during the follow-up period; (3) patients with missing data during the follow-up period.

Protocol of laser acupuncture and sham control

We selected identical acupoints in both the treatment and sham control groups. Patients in the LA group sequentially received laser stimulation energy of 4.5 J for 30 s at each of the following acupoints: bilateral Cuanzhu (BL2), Fengchi (GB20), Taiyang (EX-HN5), Shuaigu (GB8), Hegu (LI4), Taichong (LR3) and midline Yintang (EX-HN3) (Fig. 2).

LA is performed with the LaserPen-Expert 511 A (Reimers & Janssen GmbH, Germany). This device is a Gallium-Arsenide-Aluminium (Ga-As-Al) type with an output wavelength of 810 nm and an output power density of 5 W/cm². The settings and technique for laser

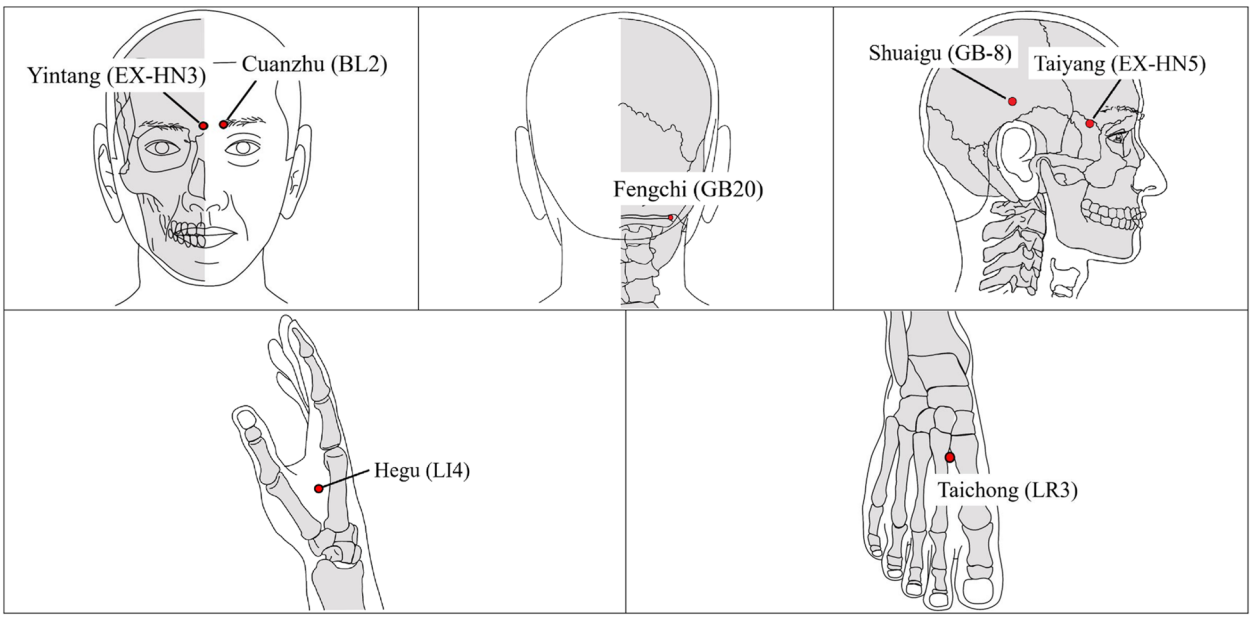


Fig. 2 Location of the acupoints used in our trial

radiation used on each acupoint are as follows: output power of 150 mW, continuous mode, applied with vertical contact pressure for a duration of 30 s, providing a stimulation energy of 4.5 J. We utilized a special attachment from Reimers & Janssen GmbH, with the LaserPen-Expert. The length of the attachment is two centimeters. Which provided a pinpoint, concentrated laser beam with a diameter of four millimeters to maintain even distribution of strength and energy across all treated acupoints.

Control patients received sham treatment with no laser output and followed the same treatment protocol as the LA group. Each treatment course lasted for about 10 min, including 13 acupoints, with each acupoint being stimulated for 30 s. After the stimulation of each acupoint, there was a 20-second period for positioning the next acupoint. The experimental room utilized the standard treatment room at Taichung Veterans General Hospital. It was a confined space of approximately 10 square meters, maintained at a temperature of 20 to 28 °C and a relative humidity of 50 to 70%. The treatment protocol was conducted over a course of 8 sessions that spanned 4 weeks. During treatment, patients were asked to wear eye goggles and noise-canceling headphones to inhibit visual and auditory perception.

Selected acupoints were localized according to the WHO Standardized Acupuncture Point Location guidelines [21]. The laser application procedures were performed by the same experienced physician who was a well-trained and a licensed Chinese medicine practitioner in Taiwan. The Chinese medicine practitioners are also required to wear eye goggles during the treatment session.

Trial outcomes

The co-primary efficacy outcomes were (1) changes from baseline in monthly migraine days (MMD), and (2) changes from baseline in acute headache medications usage days per month between baseline and each of the three follow-up time points (4th, 8th and 12th weeks) after randomization.

We classified 4 secondary efficacy outcomes in our study, which were ≥30% reduction in migraine days between baseline and each follow-up time point, and changes in the following parameters: (1) headache lasting time; (2) headache severity (based on NRS) and (3) MIDAS score.

The safety outcome was defined as any adverse event (AE) that had occurred during LA or sham treatment, and during the follow-up period. Serious AEs were defined as those that resulted in patients withdrawing from the trial. We recorded any patient-reported AEs, including but not limited to nausea, dizziness, local paresthesia, abnormal cutaneous sensations of warmth, fatigue, and ecchymosis. Trial physicians also inquired patients on the incidence of any AE during treatment protocol. Trial physicians used similar questions in the end of each treatment session (e.g., "Did you experience any discomfort during the treatment?").

Treatment credibility

To evaluate the reliability of treatment blinding and ascertain the credibility of the assigned treatment groups, a credibility questionnaire was administered at each follow-up assessment. Within the questionnaire, patients were queried about their perceptions regarding whether they had undergone laser acupuncture following the principles of traditional Chinese medicine (TCM) or received sham treatment [22]. At the end of the study, participants were surveyed about the treatment logical and their likelihood of recommending the received treatment to others. The questionnaire above was validated using the Kaiser-Meyer-Olkin (KMO) criterion to assess sampling adequacy (threshold > 0.5), and Bartlett's test of sphericity to determine the suitability of the questionnaire items for factor analysis ($p < 0.05$).

Statistical analyses, missing data imputation and sample size calculation

The baseline characteristics and clinical outcomes described are based on the ITT population, which included all randomized participants. In addition, we also performed another clinical outcome analysis based on the PP population which are available in the supplement.

Continuous data were presented as mean ± standard deviation (SD), and categorical data as number and percentage, while discrete variables were expressed as median and interquartile range (IQR). We used Fisher's exact test and chi-squared test to analyze categorical variables. While discrete variable analyses were performed using the Mann-Whitney U test. We used the chi-squared test to analyze the co-primary efficacy outcomes and the secondary efficacy outcomes (except for ≥30% reduction in migraine days). We used Fisher's exact test to analyze the ≥30% reduction in migraine days between baseline and each follow-up time point, the safety outcome, the serial guess of treatment type as well as the treatment credibility.

Participants with missing data at baseline were excluded from our study. If missing data occurred during the follow-up period, we used the last observation carried forward method to impute missing data for the ITT analysis, while ensuring exclusion from PP analysis for such case.

We assumed that the LA group would show a 60% improvement and the sham group a 20% improvement.

Therefore, a sample size of 30 patients in each group was needed to yield a power of 80% with a type I error of 5%. The power estimation was based on the randomized controlled trial of LA for children with non-selected type headache with 48 participants [17] and another randomized controlled trial on traditional acupuncture for migraine prevention with 66 participants [12]. With an estimated dropout rate of 10%, we estimated that at least 30 patients are needed in each group. Totally 60 patients were needed for randomization. Since our study is a pilot trial, there have been no previous trial specifically investigating LA for CM managed pharmacologically with unsatisfactory effect. Indeed, we lack sufficient research to estimate efficacy. Therefore, we also used post hoc power analyses based on the primary efficacy, changes from baseline in MMD at 12th weeks, to assess the adequacy of our sample size. All tests were two-sided. Statistical significance was set at *P* value less than 0.05 ($p < 0.05$). All analyses were done using the statistical package SAS version 9.4 for Windows.

Results

Participants

We recruited 98 potential participants with CM managed pharmacologically at baseline. Of these, we excluded 3 patients (3.1%) had received traditional acupuncture at baseline and one patient (1.0%) with cognitive or psychological impairment. In addition, 34 patients (34.7%) refused to participate. Finally, 60 patients were eligible to participate in our study, including 30 patients received LA and 30 patients in the sham group (Fig. 1).

Of the 60 participants, their median age was 42.0 years; their median BMI at baseline was 22.9; their median duration of migraine was 10.0 years; the male/female ratio was 15/45. The baseline demographic and clinical characteristics of both groups were similar (Table 1).

Overall, participants reported an average of 19.8 migraine days per month, and 9.4 days of acute headache medications usage per month, and no difference of these two data was found between the two groups. At the time of randomization, 55 patients (91.7%) have previously received preventive treatment and all participants reported concurrent use of acute headache medications. No inter-group difference was found in terms of the kinds of preventive treatment, including topiramate ($p = 1.000$); beta-blockers ($p = 1.000$); calcium-channel blockers ($p = 0.791$) and tricyclic antidepressants ($p = 1.000$) (Table 1).

Co-primary efficacy outcomes

Table 2 and Fig. 3 compare the primary efficacy outcomes between LA and Sham groups, with results presented as changes from baseline to the 4th (treatment completed), 8th and 12th weeks.

The change from baseline in MMD revealed no statistical difference between LA and Sham groups at 4th week (3.4 vs. 1.5 days, $p = 0.190$). In the subsequent follow-up period, the LA group had a significantly greater reduction in MMD (5.2 vs. 1.5 days at 8th week, $p = 0.015$; 7.3 vs. 1.8 days at 12th week, $p = 0.001$) (Figs. 3a). LA group had a significantly greater reduction in the number of days with acute headache medications usage days per month (3.1 vs. 0.4 days at 4th week, $p = 0.007$; 3.2 vs. 0.0 days at 8th week, $p = 0.005$; 3.9 vs. 0.0 days at 12th week, $p < 0.001$) when compared with the sham group (Figs. 3b).

Regarding the PP analyses, we recruited 57 patients, including 28 patients received LA and 29 patients in the sham group (Fig. 1). The baseline demographic was revealed in supplementary Tables 1 and the co-primary outcomes were revealed in supplementary Table 2. In PP analyses, the LA group also had a significantly greater reduction in MMD (6.7 vs. 0.8 days at 8th week, $p = 0.001$; 8.7 vs. 1.1 days at 12th week, $p < 0.001$) and the number of days with acute headache medications usage days per month (3.3 vs. 0.0 days at 4th week, $p = 0.002$; 3.7 vs. 0.6 days at 8th week, $p = 0.001$; 4.3 vs. 0.7 days at 12th week, $p < 0.001$) (supplementary Table 2).

Secondary efficacy outcomes

In terms of the mean headache severity (presented as NRS), the LA group showed significant reductions, compared with sham group (1.1 vs. 0.2 at 4th week, $p = 0.011$; 1.2 vs. 0.2 days at 8th week, $p = 0.006$; 1.3 vs. -0.1 at 12th week, $p < 0.001$). Similarly, the mean headache lasting time, while showing no significant difference at 4th week and 8th week, however the difference in reduction at 12th week (2.6 vs. 0.0 h, $p = 0.006$) was significant. The MIDAS scores between the two groups were similar at 4th week, but significant reduction in LA group compared with sham group was found at 8th and 12th weeks (24.6 vs. 5.9 at 8th weeks, $p = 0.033$; 28.4 vs. 5.5 at 12th weeks, $p = 0.012$). More than 30% reduction in migraine days revealed no difference at 4th week, but such a reduction grew significantly in the LA group compared with the sham group at 8th and 12th weeks (40.0% vs. 13.3% at 8th week, $p = 0.020$; 53.3% vs. 20.0% at 12th week, $p = 0.007$) (Table 2).

Similar to ITT analyses, PP analyses also reveal that LA group had significant reductions in secondary efficacy outcomes (supplementary Table 3).

Safety outcome

Table 3 compares safety outcomes between LA and sham groups. No serious AE was observed in either group during the treatment course or follow-up period. There was

Table 1 Baseline characteristics of trial participants

	Total ($n=60$)	Group		*p* value
		Laser Acupuncture ($n=30$)	Sham Treatment ($n=30$)	
Age, median (IQR), year	42.0 (30.3–55.8)	46.0 (29.8–56.5)	41.0(30.8–55.5)	0.690[a]
Male Gender, n (%)	15 (25.0)	6 (20.0)	9 (30.0)	0.371[b]
BMI, median (IQR)	22.9 (20.6–21.7)	23.1 (20.5–27.2)	22.9 (20.7–25.6)	0.790[a]
Duration of Migraine, median (IQR), year	10.0 (5.1–20.0)	10.0 (5.8–20.0)	12.0 (4.9–20.8)	0.801[a]
Menstrual-related migraine, n (%)	27 (45)	13 (43.3)	14 (46.7)	0.795[b]
Use of preventive treatment, n (%)	55 (91.7)	28 (93.3)	27 (90.0)	1.000[c]
Topiramate, n (%)	20 (33.3)	10 (33.3)	10 (33.3)	1.000[c]
Beta-blockers, n (%)	9 (15.0)	4 (13.3)	5 (16.7)	1.000[c]
Calcium-channel blockers, n (%)	23 (38.3)	12 (40.0)	11 (36.7)	0.791[c]
Tricyclic antidepressants, n (%)	7 (11.7)	4 (13.3)	3 (10.0)	1.000[c]
Usage of acute headache medications, n (%)	60 (100)	30 (100)	30 (100)	1.000[b]
Headache characteristics, n (%)				
Nausea or Vomiting	52 (86.7)	25 (83.3)	27 (90.0)	0.706[b]
Photophobia	33 (55.0)	17 (56.7)	16 (53.3)	0.795[b]
Phonophobia	36 (60.0)	19 (63.3)	17 (56.7)	0.598[b]
Pulsating quality	39 (65.0)	19 (63.3)	20 (66.7)	0.787[b]
Unilateral location	23 (38.3)	9 (30.0)	14 (46.7)	0.184[b]
Aggravation by activity	39 (65.0)	17 (56.7)	22 (73.3)	0.176[b]
Migraine with aura, n (%)	14 (23.3)	9 (30.0)	5 (16.7)	0.222[b]
MIDAS score, mean (SD)	56.9 (47.7)	61.2 (52.9)	52.6 (42.4)	0.701[b]
Acute headache medications usage days per month, mean ± SD, day	9.4 ± 9.6	10.2 ± 9.2	8.6 ± 10.0	0.398[b]
MMD, mean ± SD, day	19.8 ± 8.9	20.8 ± 8.6	18.8 ± 9.2	0.301[b]
Headache lasting time, mean ± SD, hour	8.3 ± 7.2	8.4 ± 7.9	8.1 ± 6.5	0.605[b]
Headache severity, mean ± SD, NRS	4.3 ± 1.7	4.4 ± 1.5	4.2 ± 1.9	0.496[b]

SD standard deviation, *BMI* body mass index, *SBP* systolic blood pressure, *MIDAS* Migraine Disability Assessment, *MMD*: monthly migraine days, *NRS* Numeric Rating Scale

[a] *p* values based on Mann-Whitney U test

[b] *p* values based on chi-squared test

[c] *p* values based on Fisher's exact test

one patient in the LA group who reported a mild local paresthesia that lasted about five minutes without potential impact on quality of life, and none needed further medical intervention. The safety outcome was similar in the two groups ($p=1.000$) (Table 3). No patient withdrew from the trial during the entire course of treatment and follow-up. In addition, the safety outcome was similar between trial groups in PP analyses (supplementary Table 4).

Treatment credibility and sample size calculation

The serial guesses of treatment type, meaning participants' perceptions regarding the treatment type at each follow-up assessment, were shown in Table 4. Notably, the questionnaire encompassed five factors: Guess of treatment type at the 4th week, Guess of treatment type at the 8th week, Guess of treatment type at the 12th week, Treatment logic, and Recommendation to others. The KMO value was 0.619, and Bartlett's test yielded a *p*-value of less than 0.0001. Based on these tests, our questionnaire is feasible. At 4th week, the guess of treatment type was similar between trial groups ($p=1.000$) and most of the participants in this questionnaire chose the 'Don't know' option (66.7% in LA group vs. 66.7% in sham group). The guess of treatment type showed no significant difference between trial groups in 8th week ($p=0.358$). At the end of study, a statistically significant disparity was observed between the groups regarding patients' ability to accurately ascertain their designated allocation status, indicating that a significantly higher number of participants in the LA group accurately ascertained their designated allocation ($p=0.031$). Participants in the study assigned high and nearly identical credibility ratings to both LA

Table 2 Co-primary efficacy outcomes and secondary efficacy outcomes

	Group		*p* value
	Laser acupuncture (*n* = 30)	Sham treatment (*n* = 30)	
Co-primary efficacy outcomes			
The change from baseline in MMD, mean ± SD, day			
4th week	-3.4 ± 4.3	-1.5 ± 4.4	0.190[a]
8th week	-5.2 ± 6.6	-1.5 ± 3.8	0.015[a*]
12th week	-7.3 ± 7.0	-1.8 ± 3.5	0.001[a*]
The change from baseline in acute headache medications usage days per month, mean (SD), day			
4th week	-3.1 ± 3.9	-0.4 ± 2.6	0.007[a*]
8th week	-3.2 ± 4.3	0.0 ± 3.0	0.005[a*]
12th week	-3.9 ± 3.8	0.0 ± 2.7	< 0.001[a*]
Secondary efficacy outcomes			
The change from baseline in headache lasting time, mean ± SD, hour			
4th week	-1.2 ± 4.2	0.0 ± 3.4	0.201[a]
8th week	-2.2 ± 5.7	-0.4 ± 2.9	0.130[a]
12th week	-2.6 ± 5.3	0.0 ± 2.2	0.006[a*]
The change from baseline in headache severity, mean ± SD, NRS			
4th week	-1.1 ± 1.5	-0.2 ± 1.5	0.011[a*]
8th week	-1.2 ± 1.6	-0.2 ± 1.5	0.006[a*]
12th week	-1.3 ± 1.4	0.1 ± 1.5	< 0.001[a*]
The change from baseline in MIDAS score, mean ± SD			
4th week	-15.7 ± 23.9	-8.3 ± 24.6	0.193[a]
8th week	-24.6 ± 33.1	-5.9 ± 24.7	0.033[a*]
12th week	-28.4 ± 33.5	-5.5 ± 26.5	0.012[a*]
30% reduction in migraine days, n (%)			
4th week	8 (26.7%)	4 (13.3%)	0.197[b]
8th week	12 (40.0%)	4 (13.3%)	0.020[b*]
12th week	16 (53.3%)	6 (20.0%)	0.007[b*]

MMD monthly migraine days, *SD* standard deviation, *NRS* Numeric Rating Scale, *MIDAS* Migraine Disability Assessment

[a] *p* values based on chi-squared test

[b] *p* values based on Fisher's exact test

**p* values less than 0.05

group and sham group in the questionnaire of treatment credibility (Table 4).

The results of the post hoc power analyses which based on the primary efficacy outcome, change from baseline in MMD at 12th weeks, indicated a power of 96.6%, confirming that the sample size for this trial was deemed sufficient (Supplementary Table 5).

Discussion

Our study demonstrated that LA is a promising add-on preventive treatment for CM patients with unsatisfactory pharmacological effect. The main findings of our study are the following: (1) laser acupuncture exhibited superior, persistent, and clinically relevant benefits for migraine prevention, reducing MMD and acute headache medications usage days per month compared with the sham treatment; (2) improvements in headache severity, headache lasting time, MIDAS score and ≥ 30% reduction in migraine days were also found; (3) LA as a preventive treatment for migraine is a safe and well-tolerated therapy; (4) the efficacy of LA sustained and increased with time after treatment course for up to 8 weeks.

Our findings are consistent with the previous randomized control trial which focuses on traditional acupuncture for migraine prevention [11]. Furthermore, in our opinion, LA is not only a non-invasive acupuncture but also represents an integrated Chinese-Western treatment for migraine prevention. Although the precise mechanisms underlying the therapeutic effects of LA in treating migraine headaches remain unclear, laser acupuncture, employing low-energy laser technology, has been demonstrated by Andraus RAC et al. to enhance the activity of metalloproteinase-2 (MMP-2), with a particular emphasis on its impact on nervous tissues [23]. Additionally, according to Fernandez-Patron et al., MMP-2 is an enzyme known for its role in the degradation of extracellular matrix components. Elevated MMP-2 activity may initiate a process leading to the cleavage of CGRP into less vasoactive metabolites [24]. Consequently, a potential mechanism by which LA may alleviate migraine headaches could be attributed to the enhanced MMP-2 activity, resulting in the degradation of CGRP, and subsequently, the reduction of migraine headache. Furthermore, an interesting aspect is the sustained and increasing long-term effects of LA. Previous studies reported that LA can inhibit peripheral sensitization by decreasing the releases of pro-inflammatory neuropeptides (substance P and CGRP) [25, 26]. The therapeutic mechanisms of LA align with the pathogenesis of CM, leading to its preventive effects on migraine occurrence. From a molecular perspective, the chemical signaling molecules, which are produced from mitochondrial stimulation by LA, will lead to the activation transcription factors [27], which subsequently triggering further down-stream effects, such as increased cell proliferation, enhanced cell migration and modulation in the levels of cytokines [28]. The interactions above might explain why the effects of LA on reducing the MMD, acute headache medications usage days, headache severity, headache lasting time and ≥ 30% reduction in migraine days become increasingly evident during the subsequent follow-up period in our study (Table 2).

In TCM, the theory posits that 'qi', which is considered a vital life force or energy that flows through the body, plays a fundamental role in maintaining health and balance. The stagnation of 'qi' lead to various health issues,

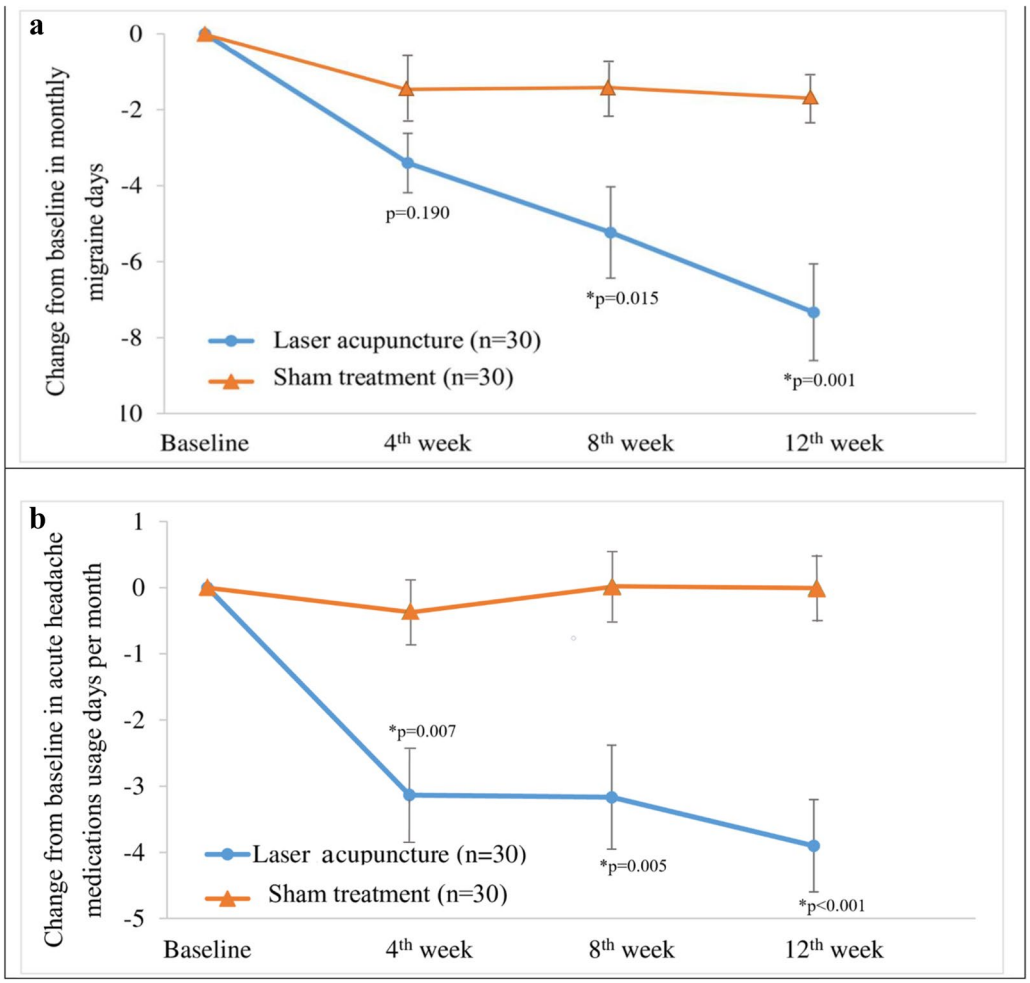


Fig. 3 Time course of the change from baseline in co-primary outcomes. **a**: Time course of the change from baseline in monthly migraine days, presented by mean ± standard error at each time point. **b**: Time course of the change from baseline in acute headache medications usage days per month, presented by mean ± standard error at each time point. * *p* values less than 0.05

Table 3 Safety outcome

	Group		*p* value
	Laser acupuncture (*n* = 30)	Sham treatment (*n* = 30)	
Any adverse event, n (%)	1 (3.3)	0 (0)	1.000[a]
Serious adverse event	0 (0)	0 (0)	1.000[a]
Mild local paresthesia	1 (3.3)	0 (0)	1.000[a]

[a] *p* values based on Fisher's exact test

including pain [29, 30]. LA, like traditional acupuncture, initiates therapeutic effects by 'de qi' [31]. 'De qi' refers to the stimulation of an acupoint, resulting in an irradiating sensation at that acupoint, which helps establish the natural flow of 'qi' [30]. By applying laser stimulation or traditional needles into acupoints, Chinese medicine practitioners aim to achieve 'de qi', thereby removing blockages and restoring the natural flow of 'qi'. Migraine headaches typically present unilaterally, initiated by the ascendant hyperactivity of liver 'yang', and often affects the Shaoyang axes, including the Gallbladder (GB) meridian. When the liver's 'yang' becomes too strong and rises upwards, it can lead to the so-called ascendant hyperactivity of liver 'yang'. Ascendant hyperactivity of liver 'yang' consequently gives rise to headache on the GB meridian in patterns of Chinese syndromes [32]. The liver and the gallbladder are exteriorly and interiorly related to each other. Both the liver and GB meridians have distribution in the head region [33]. Therefore, we selected most acupoints that could purge the liver and the gallbladder. These acupoints fall in the distribution of trigeminal and cervical dermatomes related to the trigeminal sensory pathway. Eight of the thirteen acupoints we selected (BL2, GB20, EX-HN5, EX-HN3)

Table 4 Serial guess of treatment type and treatment credibility at the end of the study

	Group		*p* value
	Laser acupuncture ($n=30$)	Sham treatment ($n=30$)	
Guess of treatment type at 4th week, n (%)			1.000[a]
Laser acupuncture	6 (20.0)	5 (16.7)	
Sham	4 (13.3)	5 (16.7)	
Don't know	20 (66.7)	20 (66.7)	
Guess of treatment type at 8th week, n (%)			0.358[a]
Laser acupuncture	8 (26.7)	4 (13.3)	
Sham	5 (16.7)	8 (26.7)	
Don't know	17 (56.7)	18 (60.0)	
Guess of treatment type at 12th week, n (%)			0.031[a*]
Laser acupuncture	17 (56.7)	7(23.3)	
Sham	4 (13.3)	7(23.3)	
Don't know	9 (30.0)	16 (53.3)	
Treatment credibility at the end of the study, n (%)			
Treatment logical	27(90.0)	25(83.3)	0.706[a]
Recommendation to others	27(90.0)	24(80.0)	0.472[a]

*p values less than 0.05

[a] p values based on Fisher's exact test

are located on the scalp, aligning with trigeminal and cervical dermatomes related to the trigeminal sensory pathway. The selection of these eight acupoints also references a published randomized clinical trial focusing on traditional acupuncture for migraine prophylaxis [12]. In addition, GB8, which is also on the scalp, is frequently used to treat migraines [11]. Furthermore, LI4 and LR3, which are effective headache-related acupoints, are included because their meridians extend to the head, helping regulate meridian qi flow. The selection of LI4 and LR3 can potentially manage local blood vessel dilation and constriction, improve blood circulation, and reduce inflammatory responses [34]. According to these mechanisms, we stimulate selected acupoints through LA to achieve 'de qi', thereby realizing the effectiveness of preventing migraine occurrence.

Based on safety assessments, 34.7% of potential participants declined due to concerns about the COVID-19 pandemic (COVID-19 pandemic outbreak in Taiwan started in March 2022). Three patients were excluded from the per-protocol (PP) analyses due to protocol violations, including two of them in the LA group and one in the sham group (Fig. 1). No patients withdrew from the study due to adverse events. Throughout the study period, only one patient reported a mild local paresthesia (mild abnormal cutaneous sensation of tingling) during the first LA treatment but did not experience any discomfort during the subsequent 7 acupuncture sessions. Therefore, we concluded LA caused only very mild adverse effects and is a safe add-on preventive treatment for CM. Our findings on safety outcomes in our patients are in line with literature in that LA is a painless, brief, and sterile treatment and facilitates stimulation of areas that are difficult or uncomfortable to needle (e.g. 'Cuanzhu' in the periocular region) [35].

Migraine patients in Taiwan often turn to TCM as an add-on non-pharmacological option after pharmacological treatments fail to provide relief. In our study, 3 patients (3.1%) had already received traditional acupuncture at baseline. Therefore, our trial specifically targeted CM patients who had not responded adequately to pharmacological management, that also implied these patients presented treatment challenges. For such patients, ≥30% reduction in migraine days often translates into a meaningful improvement in quality of life, and it is commonly used as an outcome measurement in modern pharmacological trials for CM [36, 37]. Indeed, LA showed good efficacy in reducing the headache severity in our study (Table 2). Therefore, in our study, a significant change of ≥30% reduction in migraine days in the LA group represents sufficient and meaningful clinical evidence for CM prevention.

On the other hand, LA also has an attractive role in clinical practice because of its non-invasive characteristic, leading to two noteworthy issues. First, this enables LA to serve as an add-on preventive therapy of CM. Previous studies all reported the extremely low adherence (17–25% at first year) to general preventive treatments for CM

[38, 39]. Standard preventive treatment for CM typically involves re-purposed and non-specific medications with limited safety profiles (e.g. beta-blockers and calcium-channel blockers originally for hypertension; topiramate originally for epilepsy), often encompassing various drug types, leading to poor adherence [40, 41]. Furthermore, more than two-thirds of migraineurs also reported unsatisfactory treatment responses despite various treatment approaches [9, 42]. Based on our findings, LA can be effectively applied to individuals who are undergoing or withdrawing from pharmacological preventive treatment, potentially leading to substantial improvement in migraine management. Second, such non-invasive character of LA avoids the difficulty of sham procedure design. To date, the largest randomized control trial on the efficacy of acupuncture for migraine was done in Germany. It demonstrated acupuncture was more effective in reducing migraine headaches than a waiting list control, but no statistical difference was found between acupuncture and sham acupuncture [43]. A possible explanation is among other issues, the difficulty encountered in sham acupuncture. Because in traditional acupuncture trial, sham is as invasive as true acupuncture. Other issues regarding the frequent patient-practitioner contacts, the repeated "ritual" of needling and the nonspecific physiological effects of needling. In our study, we used the activation and deactivation of laser output as the procedures for LA and sham treatment, which clearly served as an appropriate sham control trial. Furthermore, the relatively short treatment duration in our trial (10 min) was designed to minimize the patient-practitioner contacts. The results of the serial guess of treatment type and treatment credibility analyses confirm that our study achieved high-quality treatment blinding (Table 4). Interestingly, according to Table 4, the guesses regarding the treatment type at the 4th week (treatment completed) were similar between the LA and Sham groups. However, at the 8th week, the results showed a difference, though not statistically significant. By the 12th week, 56.7% of the LA group patients had correctly identified the treatment, while only 23.3% of the sham group patients had the correct answer ($p=0.031$). The findings above not only prove the reliability of our blinding but also serve as evidence of a positive and increasing response over time (during our follow-up period) to LA treatment effects for CM patients.

Our study had an innovative design focusing on LA as an add-on preventive therapy for CM. Currently, there are no trials specifically investigating LA for CM managed pharmacologically with unsatisfactory effect. We conducted a rigorous treatment protocol in each group (application of protective goggles, noise-canceling headphones, and the Chinese medicine practitioners not making additional communication with participants). We used assessments and outcome measurements as recommended in the guidelines for trials on CM [44] and applied standard acupoints, with an extremely low dropout rate. Nevertheless, our study has several limitations. First, our study design is not a double-blind trial, as only licensed Chinese medicine practitioners are permitted to administer LA in Taiwan (regulations of Taiwan Ministry of Health and Welfare). All Chinese medicine practitioners who performed LA and sham treatment are members of our study group. However, all the outcome measurements were performed by another certified case-manager, who was blinded to treatment types. Besides, in our study, each participant's LA or sham treatment was administered by the same Chinese medicine practitioner to minimize inter-practitioner variability. Second, results may not be generalizable beyond the single-center participant pool. Third, the number of participants was likely insufficient for powerful analyses. Particularly, some of the efficacy outcomes revealed only numerical differences between trial groups at 4th and 8th week. However, the result of post hoc power analyses indicate that the sample size is sufficient to substantiate the meaningfulness of the study outcomes. Fourth, we used standard acupoints instead of individualized acupoints selected to minimize the inter-individual reliability. This approach is potentially incongruent with the principles of TCM theory.

Conclusion

Laser acupuncture was effective and safe in reducing MMD and the monthly days of using acute headache medications, as well as reducing headache severity, headache lasting time and improving the quality of life for those CM patients with unsatisfactory pharmacological effects. Specifically, the efficacy of LA exhibited a progressively more pronounced impact within the subsequent 8 weeks. Based on current pilot study, conducting additional double-blind randomized controlled trials with a larger sample size, which investigating the mechanism of LA's effect on CM prevention (e.g. the direct relationship between LA and serum neuropeptides levels) and the long-term effects beyond the 12-week follow-up is a worthwhile endeavor for future research.

Abbreviations

CM	Chronic migraine
CGRP	Calcitonin gene related peptide
LA	Laser acupuncture
BMI	Body-mass index
MIDAS	Migraine Disability Assessment
NRS	Numeric Rating Scale
ITT	Intention-to-treat
PP	Per-protocol
MMD	Monthly migraine days
AE	Adverse event
TCM	Traditional Chinese medicine
MMP-2	Metalloproteinase-2
GB	Gallbladder

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12906-024-04617-9.

Supplementary Material 1.

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Authors' contributions
H.Y.W., conception and design of the study, acquisition and analysis of data, drafting a significant portion of the manuscript or figures; C.S.W., conception and design of the study, acquisition and analysis of data, drafting a significant portion of the manuscript or figures; Y.C.L., conception and design of the study, acquisition and analysis of data; C.C.C., conception and design of the study, acquisition and analysis of data; W.L.C., conception and design of the study; C.I.T., conception and design of the study; C.Y.H., acquisition and analysis of data; C.H.C.: conception and design of the study, acquisition and analysis of data, drafting a significant portion of the manuscript or figures.All authors reviewed the results and approved the final version of the manuscript.

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Availability of data and materials
All of the individual participant data collected during the trial, after deidentification at the time of immediately following publication without end date. Data are available on reasonable request from the principal investigator, Wang Chi-Sheng. Proposals should be directed to sam7227632@gmail.com. To gain access, data requestors will need to sign a data access agreement.

Declarations

Ethics approval and consent to participate
The study protocol was approved by the ethics committee of the Institutional Review Board at Taichung Veterans General Hospital (approval number: CF22082B). All participants provided written informed consent prior to enrolment.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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